

Lab Exercise — Modify the Activity-to-Activity Communication Programs

Today's lab: take the three activity-to-activity communication programs already provided — `DataTransferApp-Kotlin`, `ColorPickerApp-Kotlin`, and `StudentDetailsApp-Kotlin` — and implement the modifications below in each. Ideas are ordered roughly from easiest to most challenging; attempt at least one per program.

- 1. `DataTransferApp-Kotlin` (Intent extras — one-way transfer)**
1. **[Easy]** Add a second input field (e.g. age or favorite color) and send it as a second extra alongside the name, so `SecondActivity` displays both. Practice using multiple `putExtra()/getExtra()` calls, including a non-String type.
 2. **[Medium]** Add validation: if the name field is empty when "Send" is tapped, show a Toast message and do not start `SecondActivity`. Introduces checking input before triggering navigation.
 3. **[Challenge]** Add a third Activity: have `SecondActivity` forward the received name onward to a new `ThirdActivity` via its own Intent. Demonstrates that data does not automatically carry forward through a chain of activities — it must be explicitly re-sent at each hop.

- 2. `ColorPickerApp-Kotlin` (Activity Result API — two-way transfer)**
4. **[Easy]** Add two more color options (e.g. Purple, Orange): new buttons in the layout, a new `sendResult()` call for each, and an extended `colorFor()` when-block.
 5. **[Medium]** Add an explicit Cancel button on `SecondActivity` that calls `setResult(RESULT_CANCELED)` and `finish()` without picking a color. Currently the "cancelled" branch in `MainActivity` only triggers via the system Back button — this exercises it deliberately.
 6. **[Challenge]** Return two pieces of data from `SecondActivity` instead of one — e.g. the color name plus a light/dark shade choice — using a second `putExtra()` on the result Intent, with `MainActivity` reading both back out.

- 3. `StudentDetailsApp-Kotlin` (Parcelable — custom object transfer)**
7. **[Easy]** Add a new field to the Student data class (e.g. department). Touches every layer: the data class, the input layout, `MainActivity`'s object construction, and `SecondActivity`'s display.
 8. **[Medium]** Add logic on the receiving end: instead of just printing the raw marks value, have `SecondActivity` compute and display "Pass" or "Fail" (e.g. marks ≥ 50).
 9. **[Challenge]** Pass a list of multiple students at once (`ArrayList<Student>`) instead of a single

object, using `putParcelableArrayListExtra()` / `getParcelableArrayListExtra()`, and have `SecondActivity` display all of them.